# Do Fans Agree with Judges in the International Dance League?

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# Introduction

In recent years, social media platforms such as YouTube, Instagram and TikTok have significantly increased the viewership, exposure and global reach of dance. The International Dance League (IDL), the first professional global dance league, drew billions of views in its debut 2026 season and signed broadcast deals with ESPN2 (International Dance League, 2026c).

The league consists of six pro dance teams competing across five regular season events and a Championship. Each event has two rounds. In Round 1, six teams compete in three head-to-head matches, and in Round 2, the three Round 1 winners compete in a three-way match (International Dance League, 2026a).

In each match, a team's score combines a fan vote with scores from a six-judge panel. Each judge scores the routine on ten categories out of 10 points, for a total of 100. In Round 1, each judge awards 1 point to the team they scored higher, and 1 point is awarded to the fan vote winner. The team with the most points wins the match. In Round 2, each team's judge score is the average of the six judges' total scores. Each team's final score is its judge score plus a share of 1 point, proportional to its fan vote share. Teams are ranked first, second, and third by their final score. The Championship has seven judges and no fan vote.

Season standings are based on cumulative points earned across all events. At each event, teams earn their Round 1 match points plus 7, 5, or 3 points for finishing first, second, or third in Round 2. Standings determine Championship seeding and contribute to team payouts (International Dance League, 2026a).

Fan votes are collected through the IDL web application and are open to both in-venue attendees and YouTube livestream viewers. Voting opens after all teams in a match have performed and remains open for 60 seconds (International Dance League, 2026a).

IDL's format allows fan votes and judges' scores to be compared directly. Fans vote immediately after watching the same routines the judges score, and because every team competes at every event, these comparisons can be tracked for each team across the season. YouTube livestream viewers also leave time-stamped comments, providing a real-time record of fan reactions to each routine.

This paper measures the agreement between fan votes and judges' scores, identifies patterns in fan voting, and assesses whether YouTube livestream chat sentiment reflects fan vote share and judges' scores. It also discusses implications for teams, in how they can connect with both fans and judges, and for the league, in designing its scoring system.

## Relavant Prior Work

Major sports leagues use fan votes to select players for all-star games. Kunz-Kaltenhäuser et al. (2023) compared voting patterns for the NFL's All-Pro awards, which are selected by experts, and Pro Bowl selections before and after public balloting was introduced. The study used panel probit regression to model each selection as a function of performance markers

(e.g., touchdowns, yards, sacks) and nonperformance markers (e.g., age, previous awards, team win-loss record). In the MMA, fans can score a bout after it ends using the same 10-point scorecard as judges. Berthet (2024) used percent agreement to measure how closely judges' official scores and consensus scores aligned with these fan scorecards.

Formula E's FanBoost, used from 2014 to 2022, let fans directly influence race outcomes. Fans voted for their favourite drivers before each race, and the drivers with the most votes received a short burst of extra power during the second half of the race (e-formula.news, n.d.). Fans tended to vote for well-known drivers and championship contenders (Autosport, 2018).Formula E discontinued FanBoost in 2023 after teams and promoters unanimously agreed to remove it as part of an effort to make race outcomes more performance-based (The Race, 2022). No prior sports research has examined judge-fan agreement or fan voting patterns in a competition where fan votes directly affect match outcomes and season standings.

Sentiment analysis is widely used to study sports fans. Chang (2019) used a sentiment classifier to sort tweets posted during the Super Bowl 50 game into discrete emotion categories and found that emotion shifts aligned with scoring events. Positive emotion word counts increased after a fan's team scored, and negative emotion word counts increased after the opposing team scored. Kobs et al. (2020) built two sentiment classifiers for Twitch-specific slang and emotes and used the classifier outputs to weakly train a neural network. Applied to a livestreamed gaming convention keynote, the model showed a sharp sentiment drop when the audience learned that a newly announced game would be released for smartphones only. However, both studies interpret sentiment by linking it to events during the game or stream, such as a team scoring. Whether sentiment agrees with an independent fan vote has not been studied in a sports context.

No published research on the International Dance League currently exists.

# Data & Methods

## Dataset

Event metadata, judges' scores, and fan votes of the 2026 season were scraped from the IDL website (International Dance League, 2026d). The dataset includes 15 Round 1 matches and 5 Round 2 matches.

IDL streams on two YouTube channels, @idl.global and @steezy. YouTube live chat comments can only be accessed after a stream ends if live chat replay is enabled. Since IDL did not enable this feature for most of its livestreams, the available live chat data is limited. Live chat replays were available for four events, with 23,238 comments in total. Table 1 provides comment counts by event and channel. The Seoul event had the most comments, and its @idl.global stream had about eight times as many comments as its @steezy stream.

Table 1: YouTube Live Chat Comments by Channel

| Event | Channel | Comments |
|---|---|---|
| Sydney | @steezy | 411 |
| Seoul | @idl.global | 19775 |
| Seoul | @steezy | 2517 |
| Los Angeles | @steezy | 226 |
| Championship | @steezy | 309 |

Only comments posted during each team's performance were retained to ensure that each comment was attributed to a specific team and match. Comments posted before, between, or after performances were removed, leaving 3,989 comments.

Table 2 presents comment counts for each team at each event. Except for the Seoul event, every team received 31 or fewer comments per event, and most received fewer than 15. A team's sentiment based on so few comments would reflect only a small number of viewers, so the sentiment analysis was restricted to the Seoul event, which had between 285 and 969 comments per team.

Table 2: YouTube Live Chat Comments by Team

| Team | Sydney | Seoul | Los Angeles | Championship | Total Comments |
|---|---|---|---|---|---|
| 1MILLION | 5 | 969 | 5 | 1 | 980 |
| BROTHER-HOOD | 17 | 914 | 31 | 13 | 975 |
| ROYAL FAMILY | 5 | 610 | 1 | 0 | 616 |
| QUICK STYLE | 1 | 613 | 1 | 1 | 616 |
| JAM | 6 | 470 | 8 | 3 | 487 |
| REPUBLIC GRV | 13 | 285 | 9 | 7 | 314 |

## Sentiment Classification

IDL's YouTube live chat comments differ substantially from standard English. They include shoutouts to teams and dancers, dance-specific slang, spelling errors, and abbreviations. An existing sentiment classification model, AmaanP314/youtube-xlm-roberta-base-sentiment-multilingual (AmaanP314, 2024), is used as the baseline. The model was pre-trained on approximately 198 million multilingual tweets and fine-tuned on over 1 million YouTube comments.

A sample of 500 live chat comments was manually labelled with sentiment. The sample was drawn using stratified sampling by team to prevent over-representation of teams with higher

comment volume. Each comment was labeled as positive, negative, or neutral based on the sentiment it expressed toward the team's performance. Labeling was performed independently by two annotators familiar with dance terminology, abbreviations, and slang. The annotators agreed on 81% of comments, and only two comments received opposing labels. Limitations of this process include subjectivity in judging mixed or ambiguous comments and short exclamations.

The baseline model was evaluated on the labelled sample. It achieved an accuracy of 0.479, a macro F1 of 0.470, and a macro recall of 0.513. The baseline model was fine-tuned on the labelled sample to improve performance on IDL live chat comments. The fine-tuned model achieved an accuracy of 0.705, a mean macro F1 of 0.649, and a macro recall of 0.652. Fine-tuning on IDL-specific comments improved the model's ability to distinguish sentiment classes. The fine-tuned model was then used to label the sentiment of live chat comments from the Seoul event.

### Agreement Measure

Agreement between judges, fans, and live chat sentiment was measured with Kendall's tau-b, a rank correlation coefficient that ranges from $-1$ to 1 (Kendall, 1945). It was chosen because it handles tied values and compares only the order of values, so the measures do not need to be on the same scale. Unlike percent agreement, which only records whether two measures pick the same winner, tau-b applied to margins also measures whether larger margins under one measure correspond to larger margins under the other.

## Exploratory Data Analysis

### Fan Vote

#### Fan Vote Turnout

Fan vote counts increased at every event. Seoul's maximum match votes were more than four times those of New York. Estimated online votes were calculated by subtracting venue capacity from maximum match votes. Actual online votes may be higher. Estimated online votes also increased at every event, and in Seoul, at least 66% of votes came from online viewers. Fan vote counts for Los Angeles were not available.

Table 3: Fan Vote Counts by Event

| Event | Date | Max Match Votes | Venue Capacity | Est. Online Votes |
|---|---|---|---|---|
| New York | May 02, 2026 | 3,008 | 3500 | 0 |
| Vancouver | May 23, 2026 | 9,671 | 7000 | 2,671 |
| Sydney | Jun 20, 2026 | 10,521 | 5500 | 5,021 |
| Seoul | Aug 01, 2026 | 13,343 | 4500 | 8,843 |
| Los Angeles | Sep 19, 2026 | nan | 5000 | nan |

Fan Vote Winning Margins

In Round 1, fan vote winning margins were generally wide. Of the 15 Round 1 matches, 11 had margins above 40 percentage points, and 4 had margins of 20 points or less. Match 2 had the widest margin at four of five events.

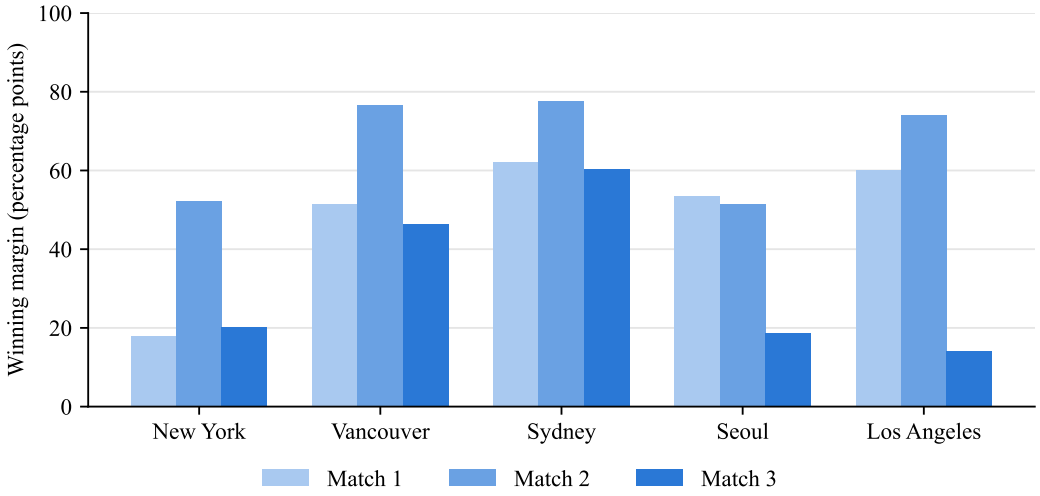



Figure 1: Round 1 Fan Vote Winning Margins

In Round 2, the margins between first and second place were wide, exceeding 40 percentage points at all five events. The margins between second and third place were narrow, ranging from 1 to 18 points.

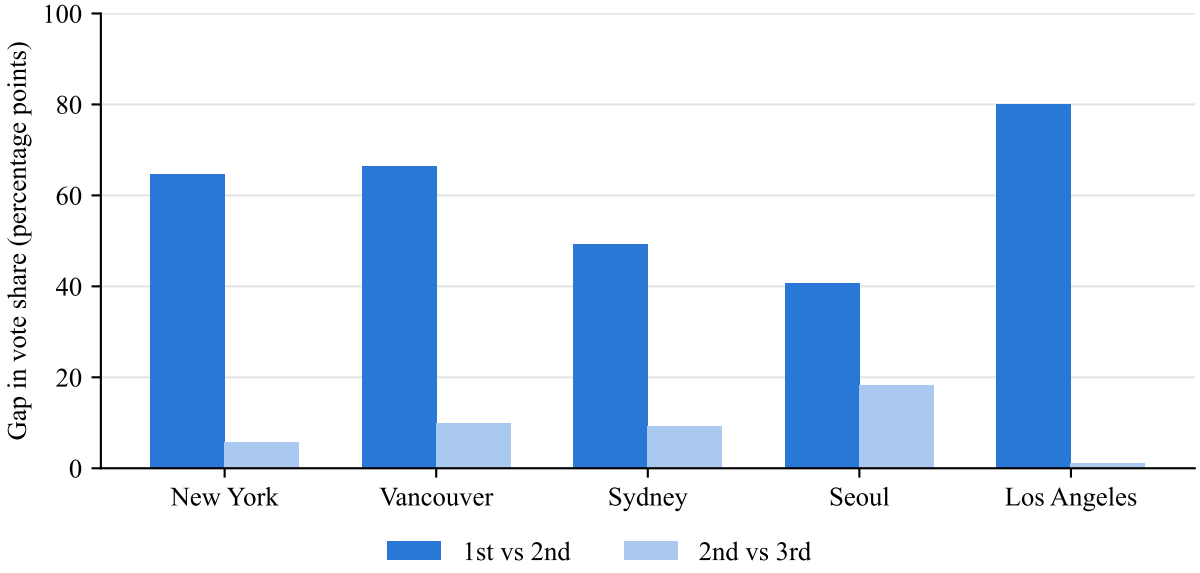



Figure 2: Round 2 Fan Vote Winning Margins

Fan Vote Share by Team

In Round 1, Brotherhood had the highest average fan vote share at 78% and won all five of its Round 1 fan votes. 1MILLION was the only other team with an average fan vote share above 50%, at 58%, and won three fan votes. Royal Family, GRV, and Jam Republic had similar averages, ranging from 44% to 48%, with two fan vote wins each. Quick Style had the lowest average share at 25% and won one fan vote.

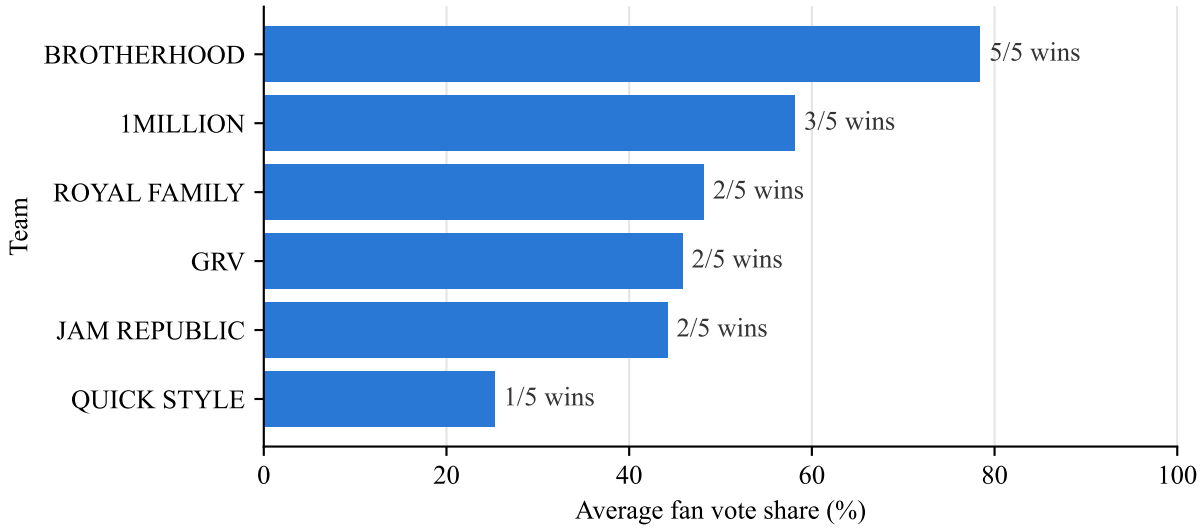



Figure 3: Average Round 1 Fan Vote Share by Team

In Round 2, Brotherhood had the highest average fan vote share at 76% and won all five of its Round 2 fan votes. No other team averaged more than 17% or won a Round 2 fan vote.

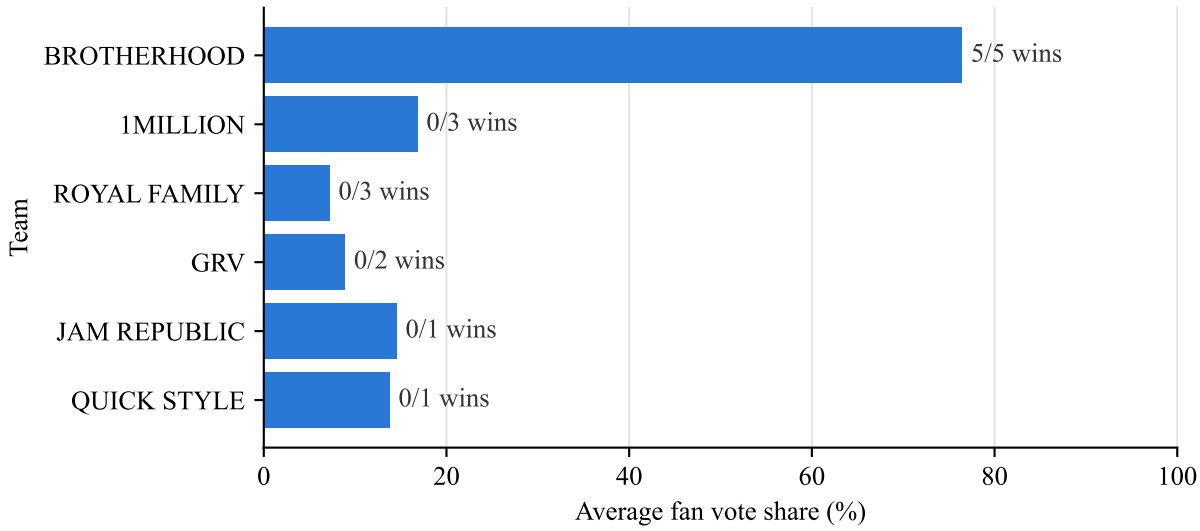



Figure 4: Average Round 2 Fan Vote Share by Team

## Judge Score

Judges' total scores were concentrated near the top of the 100-point scale. Scores ranged from 77 to 100, with a mean of 88.34 and a standard deviation of 4.40. About 68% of scores fell within one standard deviation of the mean, between 83.94 and 92.74.

Table 4: Judge Scores Summary

| N Scores | Mean | SD | Median | Min | Max |
|---|---|---|---|---|---|
| 270 | 88.34 | 4.40 | 89.00 | 77.00 | 100.00 |

Judges' category scores ranged from 5 to 10, means ranged from 8.64 to 9.47, and every category except Stamina had a median of 9.0. Stamina and Projection/Communication had the highest means, at 9.47 and 9.22. Stamina also had the lowest variability, with a standard deviation of 0.52. Cleanliness, Technical Execution + Authenticity, Choreography Complexity, Spacing, and Stylistic Athleticism had the lowest means, between 8.64 and 8.69. Choreography Complexity had the highest variability of all categories, with a standard deviation of 0.77.

Table 5: Judge Scores by Category

| Category | Mean | SD | Median | Min | Max |
|---|---|---|---|---|---|
| Stamina | 9.47 | 0.52 | 9.50 | 8.00 | 10.00 |
| Projection/Communication | 9.22 | 0.68 | 9.00 | 7.00 | 10.00 |
| Musicality | 8.85 | 0.65 | 9.00 | 7.00 | 10.00 |
| Staging | 8.76 | 0.69 | 9.00 | 7.00 | 10.00 |
| Creativity | 8.72 | 0.73 | 9.00 | 6.50 | 10.00 |
| Cleanliness | 8.69 | 0.74 | 9.00 | 6.00 | 10.00 |
| Technical Execution + Authenticity | 8.69 | 0.64 | 9.00 | 7.00 | 10.00 |
| Choreography Complexity | 8.66 | 0.77 | 9.00 | 5.00 | 10.00 |
| Spacing | 8.65 | 0.61 | 9.00 | 7.00 | 10.00 |
| Stylistic Athleticism | 8.64 | 0.74 | 9.00 | 6.50 | 10.00 |

### Judge Winning Margins

In Round 1, judge vote margins were generally wide, similar to fan vote. Of the 15 Round 1 matches, 11 were decided by unanimous 6-0 judge votes, 1 by a 5-1 split, and 3 ended in a 3-3 tie, in which the fan vote point decided the winner.

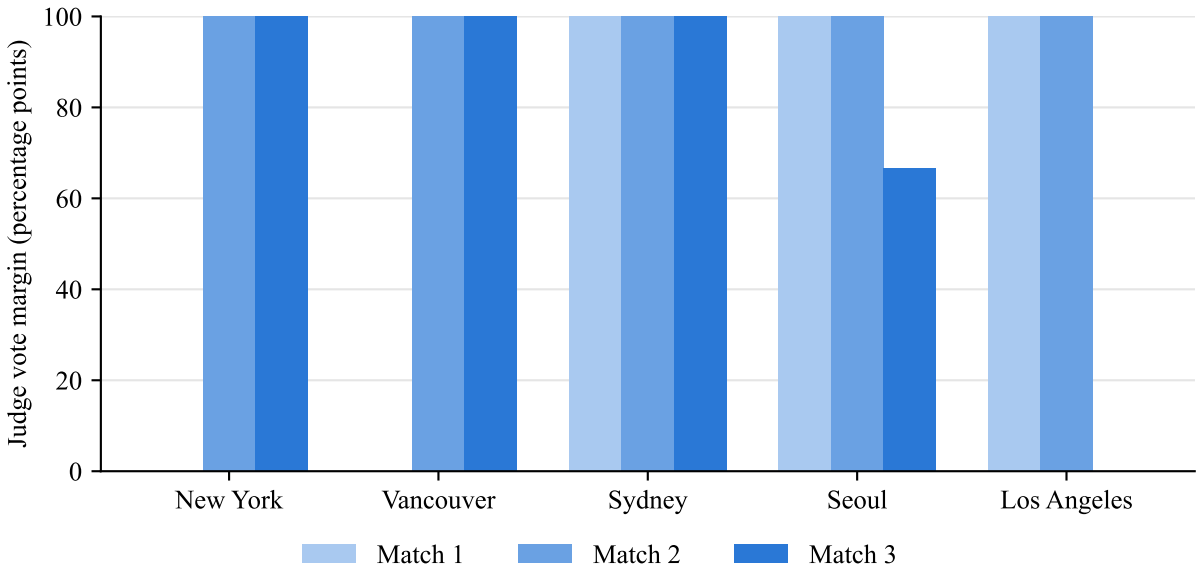



Figure 5: Round 1 Judge Vote Margins

In Round 2, the margins between first and second place ranged from 1.8 to 8.7 points in average judge score. The margins between second and third place were smaller, ranging from 0.7 to 3.7 points. In Vancouver, second and third place were separated by 0.7 points, so the fan vote, worth up to 1 point, could have changed their order. This mirrors fan vote, where first place also won by a wide margin and the gap between second and third was narrower.

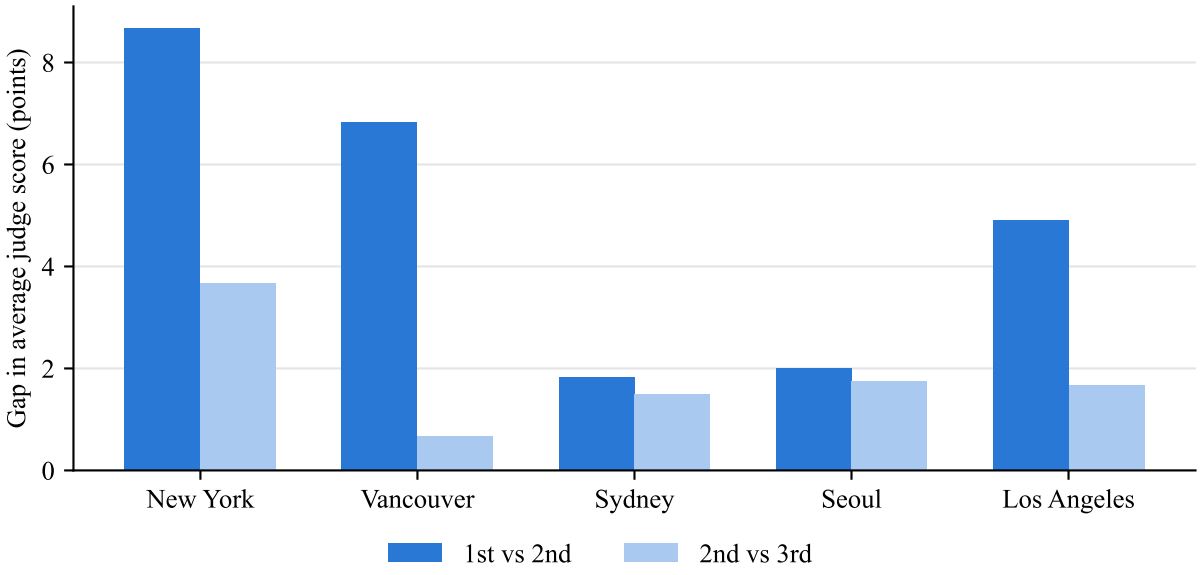



Figure 6: Round 2 Judge Score Gaps

Judge Results by Team

In Round 1, Brotherhood had the highest average judge vote share at 100% and won all five of its Round 1 judge votes. Royal Family and 1MILLION were the only other teams with an average judge vote share above 50%, at 70% and 57%, and won three and two judge votes. GRV and Jam Republic averaged 40% and 23%, with one judge vote win each. Quick Style had the lowest average share at 10% and won no judge votes.

Judges and fans agreed on the top and bottom teams: Brotherhood ranked first and Quick Style last. In the middle, judges favoured Royal Family more than fans did, while fans favoured Jam Republic more than judges did.

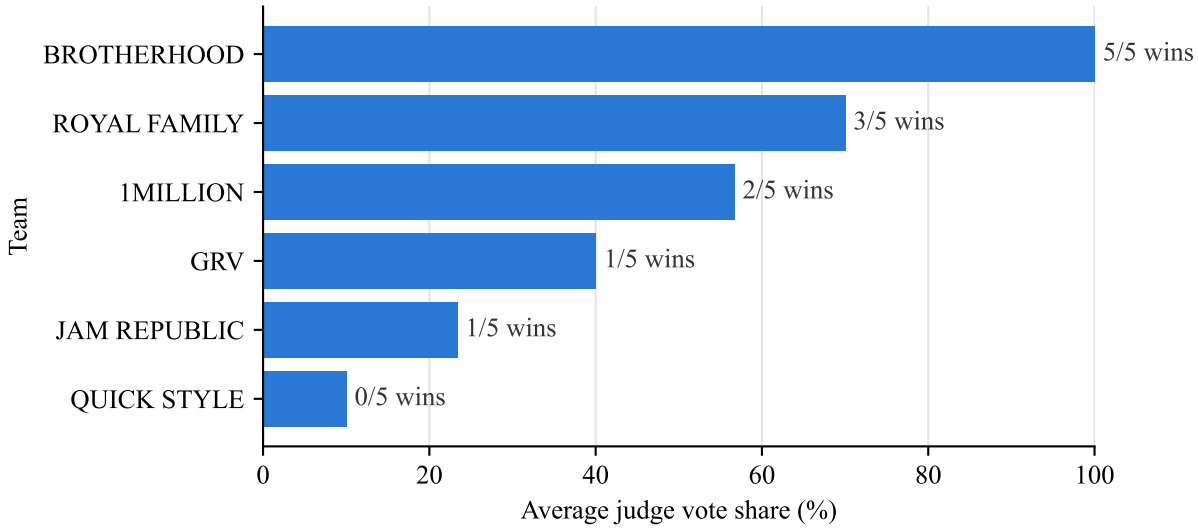



Figure 7: Average Round 1 Judge Vote Share by Team

In Round 2, Brotherhood had the highest average judge score at 94.3 and placed first in all five of its Round 2 matches, consistent with the fan vote. No other team averaged more than 90.8 or placed first in a Round 2 match.

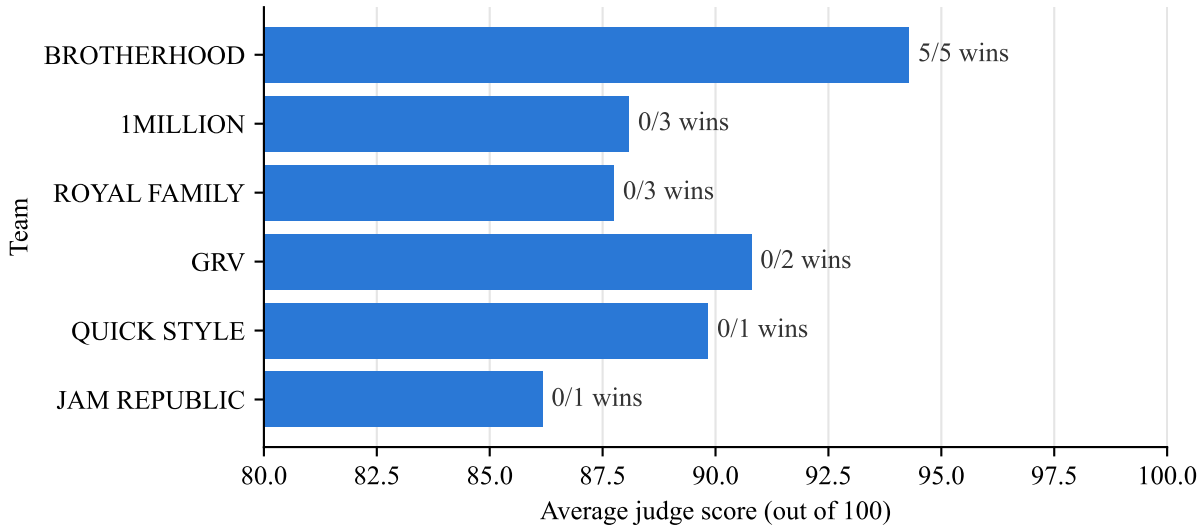



Figure 8: Average Round 2 Judge Score by Team

## Livechat Sentiment

During the Seoul event livestream, 52.1% of comments posted during routines were positive, 28.0% were neutral, and 19.9% were negative. Sentiment was similar on both channels, with @steezy slightly more positive than @idl.global.

Table 6: Live Chat Sentiment, Seoul

| Channel | Comments | Positive (%) | Neutral (%) | Negative (%) |
|---|---|---|---|---|
| @idl.global | 3,382 | 51.2 | 28.4 | 20.4 |
| @steezy | 479 | 58.5 | 25.1 | 16.5 |
| Combined | 3,861 | 52.1 | 28.0 | 19.9 |

### Livechat Sentiment Winning Margins

Net sentiment is the number of positive comments minus the number of negative comments, divided by the team's total comments during its routine, and ranges from $-1$ to 1. In Round 1, live chat net sentiment margins varied across matches, whereas judge and fan margins were mostly wide. Match 2 had the widest margin at 0.52, while Match 1 had the narrowest at 0.13. In Round 2, margins were narrow, at 0.13 between first and second place and 0.17 between second and third. Unlike the judge and fan votes, where first place won by a wide margin, live chat sentiment did not clearly separate the Round 2 winner.

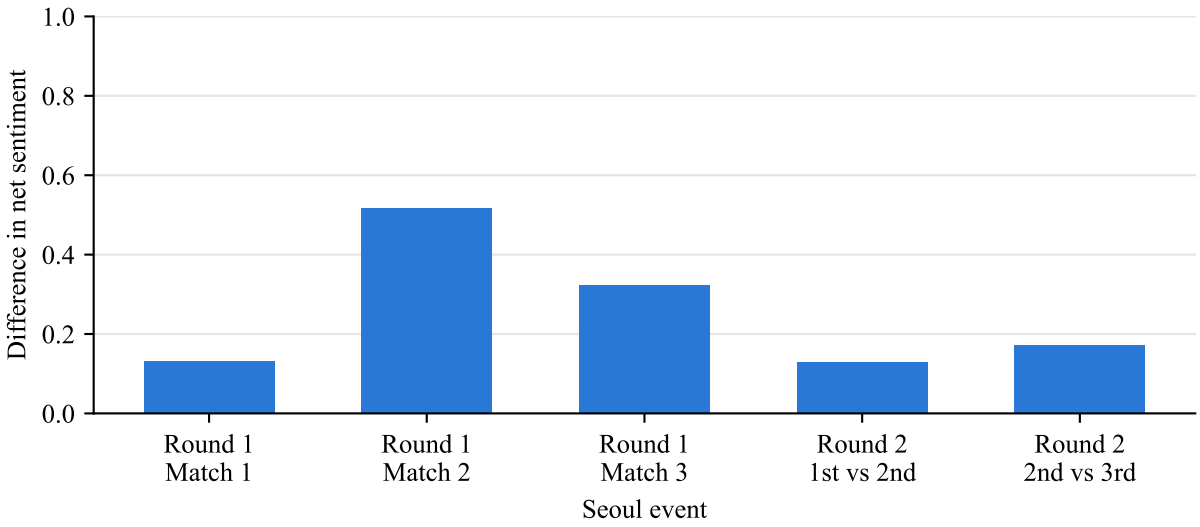



Figure 9: Live Chat Net Sentiment Margins, Seoul

# Results

Judges and fans chose the same winner in 11 of 12 untied Round 1 matches and all five Round 2 matches. In the Seoul event, live chat sentiment matched the fan and judge winner in all four matches but reversed second and third place in Round 2.

## Agreement Between Fan Votes and Judges' Scores

Judge margins and fan margins showed strong positive agreement in both rounds. In Round 1, tau was 0.73, with a 95% confidence interval from 0.51 to 0.87 and a p-value of .002, across 15 matches. In Round 2, tau was 0.66, with a 95% confidence interval from 0.48 to 0.87 and a p-value of .002, across 5 matches. Teams that won by larger margins with the judges generally also won by larger margins with fans.

All the judging categories showed positive agreement with fan margins in both rounds. Projection/Communication and Creativity showed the strongest agreement with fan margins in both rounds. Stylistic Athleticism showed strong agreement in Round 1 and moderate agreement in Round 2, whereas Musicality showed weak agreement in Round 1 and strong agreement in Round 2.

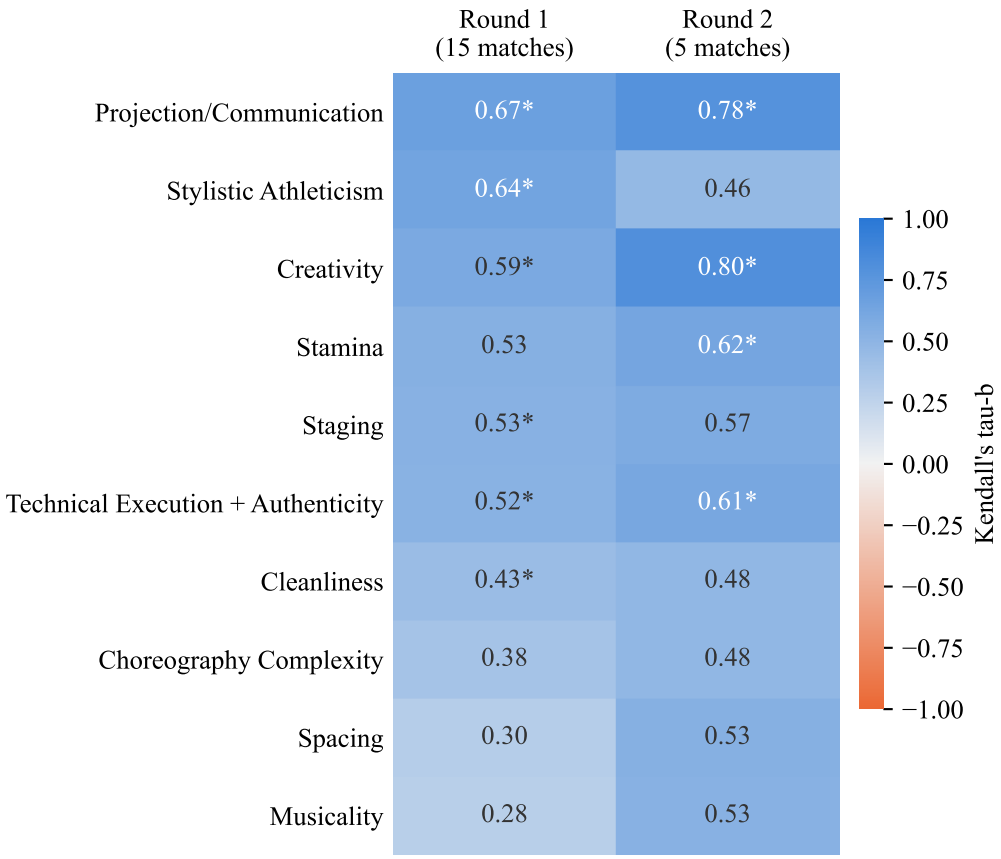



Figure 10: Agreement Between Judging Categories and Fan Margins by Round

### Agreement Between Fan Votes and Livechat Sentiment

At the Seoul event, live chat sentiment margins and fan margins showed positive agreement in Round 1, with a tau of 0.47, a 95% confidence interval from 0.38 to 1.00, and a p-value of .50. In Round 2, tau was 0.33 with a p-value of 1.00. With three Round 1 matches and one Round 2 match, these results are not statistically significant.

### Agreement Between Judges' Scores and Livechat Sentiment

Live chat sentiment margins and judge margins showed positive agreement in Round 1 at Seoul, with a tau of 0.64, a 95% confidence interval from 0.38 to 1.00, and a p-value of .50. In Round 2, tau was 0.33 with a p-value of 1.00. These results are not statistically significant.

## Conclusion

Judges and fans showed strong positive, statistically significant agreement in both rounds of the 2026 IDL regular season. Teams that won by larger margins with the judges generally won by larger margins with fans. Projection/Communication and Creativity showed the

strongest agreement with fan margins in both rounds, suggesting that these are the categories where teams' performances connect most with both judges and fans.

Fans determined the winner only in the three Round 1 matches where the judges tied. Fans largely agreed with the judges and never overturned a judge decision. This suggests that fans and judges value similar qualities in a performance. However, the agreement could also occur if fans voted for popular teams that also happened to perform well.

At the Seoul event, live chat sentiment showed positive agreement with both fans and judges and identified the same winner in all four matches. These associations were not statistically significant. Limitations include that only 17.2% of comments were posted during performances, and commenters are a subset of online viewers and do not include in-person voters.

## Future Extensions

Future extensions could test whether fan votes reflect performance or popularity by modeling fan vote share with judge scores and popularity measures, such as social media following. Additional seasons would also make it possible to test whether fans keep voting for a team they support when judges favor its opponent. Enabling live chat replay for all future livestreams and collecting comments from other IDL platforms would provide enough matches to properly test live chat sentiment as a proxy for fan votes.

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